Line-preserving deformations of Fano flows: span, quadratic obstructions, and structural limits

AI-assisted research draft for independent human verification
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AI-assistance disclosure. OpenAI Codex agents (GPT-5-series models, accessed in July 2026) developed the fixed-line formulation, found the telescoping potential, derived the two-cycle normal form, the Petersen example, and the cotree–diamond iteration, constructed the cyclically four-edge-connected order-60 example, wrote the search and checking programs, and drafted this manuscript under human direction. The checkers and the present proof exposition are themselves AI-assisted; "independent" below means logically separate code, not independent human verification. No human author or referee is represented as having certified the results. This draft must not be submitted until a human author has checked every proof and computation, verified the literature search, and accepted ordinary scholarly responsibility. The results are partial structural statements and do not prove or disprove the five-cycle double cover conjecture.

Abstract

Let f be a nowhere-zero $\mathbb{F}_2^3$-flow on a finite loopless cubic graph and fix a nonzero functional μ. Its kernel determines a Fano line L, a spanning factor F_L, and a binary vector recording the components of F_L whose four affine boundary colours all occur oddly. For each of the three nonzero values $t \in L$, binary cycles avoiding the value class $f^{-1}(t)$ give an initially valid line-preserving switch map. We prove that the sum of the images of these three linear maps always contains the defect vector. The proof is a short cut–cycle duality argument closed by an explicit quadratic potential which telescopes against flow conservation.

The span identity is only a first-order statement. We give an exact nonlinear normal form: flows with the same μ-projection are parametrised by two binary cycles p, q, and line cleaning is equivalent to their covering the kernel factor together with quadratic product-parity equations on its components. This yields a general obstruction when the kernel factor is a perfect matching whose simple contraction Q satisfies $|E(Q)| < 4\ \mathrm{girth}(Q)$.

We also prove that no bounded-component kernel-line normal form can be universal. Iterated cotree–diamond expansion gives simple bridgeless cubic graphs G_d, all with explicit five-cycle double covers, such that every Fano line in every nowhere-zero $\mathbb{F}_2^3$-flow has at least $d+1$ components. Explicit clean flows attain equality.

Finally, we display a nowhere-zero $\mathbb{F}_2^3$-flow on the Petersen graph with two prescribed lines that cannot be cleaned while their functional projections are fixed. The obstruction has a finite human-checkable proof: in both cases the matching contraction is K_5, and a clean lift would force a Tait colouring of the Petersen graph. The same graph has an explicit five-cycle double cover, so the example is not a counterexample to that conjecture. The manuscript and all calculations are AI-assisted and await independent human review.

More strongly, we construct a simple nonplanar cubic graph of order 60 and cyclic edge-connectivity at least four with a fixed nowhere-zero $\mathbb{F}_2^3$-flow for which all seven functional projections are uncleanable. Two completion-sound four-pole profiles cover the seven functionals, and a solver-independent checker proves the finite local classifications by exhaustive binary-cycle

enumeration and Gaussian elimination. The same graph has an explicit five-cycle double cover. Thus even in the usual reduced domain, the flow must be selected existentially; the result is not a counterexample to the FiveCDC conjecture.

1 Scope and background

All graphs in the main results are finite, undirected, loopless, and cubic; parallel edges are allowed unless simplicity is stated. Since the coefficient groups have exponent two, orientations play no role. An $\mathbb{F}_2^k$-flow is a map $f : E(G) \to \mathbb{F}_2^k$ satisfying

$$\bigoplus_{e \ni v} f(e) = 0 \qquad (v \in V(G)).$$

It is *nowhere-zero* if $f(e) \neq 0$ on every edge. At a cubic vertex the three values of a nowhere-zero $\mathbb{F}_2^3$-flow are the three points of a line of the Fano plane.

Fano colourings of cubic graphs and their relation to bridgelessness, perfect matchings, and the Fulkerson and Fan–Raspaud conjectures are well established [2, 8, 7, 5, 6]. In particular, the existence of nowhere-zero $\mathbb{F}_2^3$-flows on bridgeless graphs follows from Jaeger's 8-flow theorem [4]. This paper does not claim any of that background as new.

Our question is narrower. We fix one nonzero functional $\mu : \mathbb{F}_2^3 \to \mathbb{F}_2$, preserve the binary flow $\mu \circ f$, and ask whether the corresponding kernel line can be made *clean*, in the sense defined below. This is not the established k-line Fano-flow problem, which counts the vertex lines used by a flow. It is also different from recent work on reconfiguring arbitrary nowhere-zero flows by cycle-supported steps [1].

There is a relevant recent neighbour. Mkrtchyan studies non-conflicting $\mathbb{F}_2^2$-flows on contractions of complementary 2-factors and proves that the Petersen graph is an obstruction [9, Proposition 2.21]. Our clean-lift condition, fixed functional projection, combined-image identity, and quadratic normal form are different. Thus the general fact that "Petersen is an obstruction to a constrained flow extension" is not new; only the specific statements proved here are provisionally claimed as new.

Hušek and Šámal give an exact component-parity criterion for a five-cycle double cover derived from a nowhere-zero $\mathbb{F}_2^3$-flow [3, Theorem 3.16]. Their Conjecture 3.19 asks existentially for a flow and a functional satisfying that criterion, and is equivalent to FiveCDC in their framework. Our normal form below recovers the fixed-projection lifting equations. The order-60 construction in Section 8 shows that the existential quantifier over flows cannot be replaced by "for every flow", even under cyclic four-edge-connectivity.

For further separation from the motivating conjecture, a five-cycle double cover (5-CDC) is a list of five even edge-subsets in which every edge appears exactly twice. Král' et al. give the equivalent Desargues-configuration colouring formulation [7, Theorem 7.1]. Nothing below proves that every bridgeless graph has such a cover. Section 7 gives a 5-CDC of our Petersen example explicitly.

2 A fixed line and its switch images

Let $f : E(G) \to \mathbb{F}_2^3 - \{0\}$ be a flow and fix a nonzero functional μ. Write

$$K = \ker \mu, \qquad L = K - \{0\}, \qquad A = \{a \in \mathbb{F}_2^3 : \mu(a) = 1\}.$$

The three-element set L is a Fano line and A is its affine complement. Put

$$F_L = f^{-1}(L).$$

At each cubic vertex, an even number of incident values lie in A. Consequently every vertex has degree one or three in the spanning subgraph F_L. Let $\mathcal{K}$ be its set of connected components.

For a component $W \in \mathcal{K}$, flow conservation across its shore gives

$$\bigoplus_{e \in \delta(W)} f(e) = 0.$$

The boundary edges all have values in A. The parities of the four affine value multiplicities on $\delta(W)$ are equal. One quick proof is to write $A = b + K$: the kernel of the 3×4 matrix whose columns are $(1, k)$, $k \in K$, is generated by the all-one vector. Define the *rainbow-defect vector*

$$r \in \mathbb{F}_2^{\mathcal{K}}, \qquad r_W = 1 \quad \Longleftrightarrow \quad \text{each of the four values in } A \text{ occurs oddly on } \delta(W).$$

We call L *clean* if $r = 0$.

For $t \in L$, translation by t partitions A into two two-element orbits. Choose one orbit P_t, and put

$$M_t = f^{-1}(t), \qquad Z_t = Z_1(G - M_t; \mathbb{F}_2).$$

Thus Z_t is the binary cycle space of the graph obtained by deleting the t-valued edges. For $C \in Z_t$, set

$$\tau_t(C)_W = \left|C \cap \delta(W) \cap f^{-1}(P_t)\right| \pmod 2. \tag{2.1}$$

The complementary orbit gives the same bit, because a binary cycle crosses every cut evenly.

Adding t to the values on C preserves flow conservation. Because $C \cap M_t = \varnothing$, it creates no zero edge. Formula (2.1) is the exact defect update for this single switch, computed at the displayed initial flow. Define the combined image

$$\mathcal{W}_L = \sum_{t \in L} \operatorname{im} \tau_t. \tag{2.2}$$

3 The combined-image theorem

Theorem 3.1 (combined-line span)**.** *For every finite loopless cubic G, every nowhere-zero $\mathbb{F}_2^3$-flow f, and every nonzero functional μ,*

$$r \in \mathcal{W}_L. \tag{3.1}$$

Proof. Suppose that (3.1) fails. By finite-dimensional linear duality there is $y \in \mathbb{F}_2^{\mathcal{K}}$ such that

$$y \cdot r = 1, \qquad y \cdot \tau_t(C) = 0 \quad (t \in L,\ C \in Z_t). \tag{3.2}$$

Let $\mathcal{Y} = \operatorname{supp} y$ and

$$U = \bigcup_{W \in \mathcal{Y}} V(W).$$

For each t, equation (2.1) gives

$$y \cdot \tau_t(C) = \left|C \cap \delta(U) \cap f^{-1}(P_t)\right| \pmod 2. \tag{3.3}$$

In any graph, the cut space is the orthogonal complement of the cycle space. Hence

$$R_t = \delta(U) \cap f^{-1}(P_t) \tag{3.4}$$

is a cut of $G - M_t$. Choose a vertex indicator $x_t : V(G) \to \mathbb{F}_2$ for this cut.

Apply an invertible change of coordinates in $\mathbb{F}_2^3$ and label vectors by binary integers with

$$1 = (1,0,0), \quad 2 = (0,1,0), \quad 4 = (0,0,1).$$

We may assume

$$L = \{1,2,3\}, \qquad A = \{4,5,6,7\}, \qquad P_t = \{4, 4+t\}. \tag{3.5}$$

Let u be the indicator of U, and at each vertex put

$$q = (u, x_1, x_2, x_3). \tag{3.6}$$

Across an edge, (3.4)–(3.5) allow the zero difference and the following single nonzero difference:

$f(e)$	$q_v + q_w$
1	$(0;100)$
2	$(0;010)$
3	$(0;001)$
4	$(1;111)$
5	$(1;100)$
6	$(1;010)$
7	$(1;001)$

(3.7)

For example, a line-valued edge has both ends in the same component of F_L, so its u-difference is zero. If its value is t, only the cut x_t is unconstrained because that edge was deleted from $G - M_t$. On an affine edge, (3.4) says that the x_t-difference is the u-difference precisely when its colour lies in P_t. This proves every row of (3.7).

Define the potential

$$\mathcal{A}(u, x_1, x_2, x_3) = \big(u(x_2 + x_3),\ u(x_1 + x_3),\ x_1x_2 + x_1x_3 + x_2x_3\big). \tag{3.8}$$

All operations are in $\mathbb{F}_2$. Direct substitution in the seven rows of (3.7) gives

$$\big(\mathcal{A}(q_v) + \mathcal{A}(q_w)\big) \cdot f(vw) = \begin{cases} 1, & f(vw) = 4 \text{ and } u(v) + u(w) = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{3.9}$$

For transparency, the seven nonzero cases reduce as follows:

$f(e)$	$q_v + q_w$	$(\mathcal{A}(q_v) + \mathcal{A}(q_w)) \cdot f(e)$
1	$(0;100)$	0
2	$(0;010)$	0
3	$(0;001)$	0
4	$(1;111)$	1
5	$(1;100)$	0
6	$(1;010)$	0
7	$(1;001)$	0

The zero-difference cases are immediate. As one sample expansion, in the colour-4 row all three x_i and u toggle; the change of $x_1x_2 + x_1x_3 + x_2x_3$ is 1, and $4 = (0,0,1)$. In the colour-5 row the changes of the first and third coordinates of $\mathcal{A}$ are both $x_2 + x_3$, and $5 = (1,0,1)$, so they cancel. The remaining rows follow by the same two-line expansion.

Sum (3.9) over all edges and regroup the left side at vertices:

$$\sum_{vw\in E(G)} (\mathcal{A}(q_v) + \mathcal{A}(q_w)) \cdot f(vw) = \sum_{v\in V(G)} \mathcal{A}(q_v) \cdot \left(\bigoplus_{e\ni v} f(e) \right) \tag{3.10}$$
$$= 0$$

by flow conservation. The right side of (3.9) sums to

$$|\delta(U) \cap f^{-1}(4)| \pmod 2. \tag{3.11}$$

On the other hand, summing the colour-4 boundary parity over the components selected by y cancels affine edges joining two selected components and leaves exactly (3.11). By the definition of r, this is $y \cdot r = 1$, contradicting (3.10). □

Corollary 3.2 (at most two components)**.** *If F_L has at most two components, then L is clean or one valid line-preserving binary-cycle switch makes it clean.*

Proof. Every vector in every $\operatorname{im}\tau_t$ has even Hamming weight: after summing (2.1) over all components, every counted affine edge is counted at both ends. If F_L has one component, its even-weight subspace is zero, so Theorem 3.1 gives $r = 0$. If it has two components, that subspace is one-dimensional. When $r \neq 0$, some $\operatorname{im}\tau_t$ in (2.2) must be nonzero and therefore contains r. One cycle producing r cleans the line. A single cycle has no overlap correction and avoids M_t, so the switch remains nowhere-zero. □

4 The exact quadratic normal form

Theorem 3.1 combines three maps computed at the initial flow. It does not assert that the corresponding three switches can be performed simultaneously. An affine edge used by two switches has a quadratic correction, and two switches may also turn a line-valued edge into zero. We now describe the exact problem without switch-order ambiguity.

Let

$$h = \mu \circ f, \qquad F_\mu = h^{-1}(0).$$

Thus $F_\mu = F_L$. Choose $b \in \mathbb{F}_2^3$ with $\mu(b) = 1$ and identify $K = \ker\mu$ with $\mathbb{F}_2^2$.

Proposition 4.1 (two-cycle normal form)**.** *The flows f' satisfying $\mu \circ f' = h$ are exactly*

$$f'_e = b\, h_e + (p_e, q_e), \tag{4.1}$$

where $p, q \in Z_1(G; \mathbb{F}_2)$. Such a flow is nowhere-zero exactly when

$$p_e \vee q_e = 1 \qquad (e \in F_\mu). \tag{4.2}$$

It is clean on every component W of F_μ exactly when

$$\sum_{e\in\delta(W)} p_e q_e = 0 \qquad (W \in \mathcal{K}). \tag{4.3}$$

Proof. If $\mu \circ f' = h$, then

$$s = f' + b\,h$$

is a K-valued flow. Its two coordinates are binary cycles p, q, which proves existence and uniqueness in (4.1). Conversely, any two binary cycles in (4.1) give a flow with projection h.

On an edge with $h_e = 1$, the value $b + (p_e, q_e)$ is automatically nonzero because it lies outside K. On an edge of F_μ, it is (p_e, q_e), proving (4.2).

The boundary of a component W consists of affine edges. Its size is even, and each binary cycle has even intersection with its cut. The parity of the affine colour b, whose residual coordinate is $(0, 0)$, is

$$\begin{aligned} \sum_{e\in\delta(W)} (1+p_e)(1+q_e) &= |\delta(W)| + \sum_{\delta(W)} p_e + \sum_{\delta(W)} q_e + \sum_{\delta(W)} p_e q_e \\ &= \sum_{e\in\delta(W)} p_e q_e. \end{aligned} \tag{4.4}$$

As in Section 2, flow conservation makes the four affine colour parities equal. Thus they are all even exactly when (4.3) holds. □

Remark 4.2. Equations (4.2)–(4.3) say that p, q cover the kernel factor and that, after its components are contracted, the affine edge set $\{e : p_e = q_e = 1\}$ has even degree at every contracted vertex. The cover clauses are monotone, but the boundary clauses contain the quadratic products $p_e q_e$. Theorem 3.1 removes the corresponding first-order obstruction; it does not solve these quadratic equations.

Corollary 4.3 (Tait graphs have no fixed-projection obstruction)**.** *Let G be a Tait-colourable cubic graph. For every binary cycle $h \in Z_1(G; \mathbb{F}_2)$, the two-cycle conditions* (4.2)–(4.3) *are feasible for $F_\mu = h^{-1}(0)$. Consequently every functional projection has a clean nowhere-zero $\mathbb{F}_2^3$-lift.*

Proof. Fix a nowhere-zero $\mathbb{F}_2^2$-flow $s = (p, q)$ supplied by a Tait colouring. Since s is nowhere-zero on every edge, p, q cover F_μ, proving (4.2).

Let W be a component of F_μ. Every edge of $\delta(W)$ has $h_e = 1$, and the binary cycle h crosses the cut evenly, so $|\delta(W)|$ is even. Let a, b, c be the parities on this cut of the three nonzero s-values $(1, 0), (0, 1), (1, 1)$, respectively. Flow conservation across the cut gives

$$a + c = 0, \qquad b + c = 0,$$

so $a = b = c$. But

$$0 = |\delta(W)| = a + b + c = c \qquad \text{in } \mathbb{F}_2.$$

Thus the $(1, 1)$-colour occurs evenly on $\delta(W)$, which is exactly $\sum_{e\in\delta(W)} p_e q_e = 0$. This proves (4.3), and Proposition 4.1 supplies the clean lift. □

5 No bounded-component kernel-line normal form

The preceding at-most-two-component corollary is useful but cannot be turned into a universal proof by replacing two with any fixed constant.

Let H be a simple cubic graph, let T be a spanning tree, and replace every cotree edge uv by four private vertices a, b, c, d and the seven edges

$$ua,\ ac,\ ad,\ bc,\ bd,\ cd,\ bv. \tag{5.1}$$

The private vertices induce the diamond $K_4 - ab$. Denote the resulting graph by $D_T(H)$.

Lemma 5.1 (component growth)**.** *Suppose H is not Tait-colourable. Every nowhere-zero $\mathbb{F}_2^3$-flow f' on $D_T(H)$ suppresses to a nowhere-zero flow f on H, and for every Fano line L,*

$$\kappa_L(f') \geq \kappa_L(f) + 1, \tag{5.2}$$

where κ_L is the number of components of the subgraph induced by the L-valued edges.

Proof. Conservation over the four private vertices of a diamond says that its two boundary values are equal. Use their nonzero common value on the suppressed edge. Conservation at the old vertices is preserved, giving the flow f.

Let $h : \mathbb{F}_2^3 \to \mathbb{F}_2$ have kernel $L \cup \{0\}$. The support of $h \circ f$ is an even edge-subset of H. It is nonempty: otherwise all values of f belong to L, and at every cubic vertex they are the three distinct nonzero elements of L, which is a Tait colouring. A nonempty even edge-subset cannot be contained in the tree T, so some cotree edge has suppressed value outside L.

In the corresponding diamond both boundary edges are outside L. The four private vertices nevertheless have positive odd degree in the line subgraph, so their line-valued edges contain at least one component isolated from the rest of $D_T(H)$.

Replacing a line-valued edge cannot merge two old line components: every line-valued path between old vertices projects, after suppressing diamonds, to a walk of line-valued edges in H. Thus the components containing old vertices number at least $\kappa_L(f)$, and the isolated diamond component supplies one more. This proves (5.2). □

Theorem 5.2 (unbounded kernel-line complexity)**.** *There are simple bridgeless cubic graphs G_d, $d \geq 0$, each having a standard five-cycle double cover, such that every nowhere-zero $\mathbb{F}_2^3$-flow f and every Fano line L satisfy*

$$\kappa_L(f) \geq d + 1. \tag{5.3}$$

Equality is attained by a flow in which the displayed line is clean. Their orders begin

$$10, 34, 106, 322, \ldots$$

and satisfy $n_{d+1} = 3n_d + 4$.

Proof. Let G_0 be the Petersen graph. Given G_d, choose a spanning tree T_d and put $G_{d+1} = D_{T_d}(G_d)$. The construction preserves simplicity and cubicity. It preserves bridgelessness because every boundary edge lies on a lifted old circuit and every internal diamond edge lies on a triangle. It also preserves non-Tait-colourability: suppressing the equal boundary colours of a hypothetical Tait colouring would colour the preceding graph. Lemma 5.1, starting from the trivial bound one, proves (5.3).

A five-cycle double cover lifts locally. If a replaced edge has two-coordinate label $\ell = \{i, j\}$, choose $k \notin \ell$, put $r = \{i, k\}$, $s = \{j, k\}$, and label the seven edges in (5.1)

$$\ell, r, s, s, r, \ell, \ell. \tag{5.4}$$

The three incident labels at every private vertex have empty symmetric difference, proving both parity and exact double coverage. Petersen has the explicit cover displayed later in (7.6), after relabelling its edges, so every G_d has a cover.

For sharpness fix $L = \{1, 2, 3\}$. In the Petersen edge order

$$(0,1), (1,2), (2,3), (3,4), (0,4), (0,5), (1,6), (2,7),$$
$$(3,8), (4,9), (5,7), (7,9), (6,9), (6,8), (5,8),$$

the flow

$$5, 1, 3, 1, 2, 7, 4, 2, 2, 3, 3, 1, 2, 6, 4 \tag{5.5}$$

has a connected L-subgraph and one circuit as its outside-line support. At each stage choose T_d to contain all but one edge of that circuit; the remaining path extends to a spanning tree.

For a line value, lift the diamond entirely inside L. For an outside value x, choose distinct outside values $y, z \neq x$ and use

$$x, y, x + y, z, x + z, y + z, x. \tag{5.6}$$

The outside support replaces one circuit edge by a four-edge path and remains a circuit. The three line-valued internal edges in that diamond form one isolated star, while all line-valued diamonds connect their old ends. The new star has the two equal affine boundary values x, x, so it is clean, and every old component retains its affine boundary parities. The connected Petersen line is clean. Exactly one clean line component is therefore added at every stage, giving equality in (5.3) with zero defect.

Finally, a cubic graph on n vertices has $n/2 + 1$ cotree edges. Each adds four vertices, so $n_{d+1} = n_d + 4(n_d/2 + 1) = 3n_d + 4$. □

6 A structural matching-contraction obstruction

Recall that a cubic graph is *Tait-colourable* if it has a proper three-edge-colouring. Equivalently, it admits a nowhere-zero $\mathbb{F}_2^2$-flow.

Theorem 6.1 (simple-contraction obstruction)**.** *Suppose F_μ is a perfect matching of a cubic graph G, and let $Q = G/F_\mu$. If*

$$Q \textit{ is simple}, \qquad |E(Q)| < 4 \operatorname{girth}(Q), \qquad G \textit{ is not Tait-colourable}, \tag{6.1}$$

then no clean nowhere-zero flow f' can have $\mu \circ f' = \mu \circ f$.

Proof. Assume that a clean lift f' exists. The edges outside F_μ become the edges of Q. Their four affine colours partition $E(Q)$. Cleanliness says that every colour class has even degree at each vertex of Q, so each is an even subgraph.

Every nonempty even subgraph of a finite simple graph contains a cycle, and therefore has at least girth(Q) edges. The strict inequality in (6.1) implies that one of the four affine colours is absent. Choose this absent colour as the base b in (4.1), and write

$$f' = b\,h + s, \qquad s : E(G) \to K.$$

On an affine edge, $s \neq 0$, since $f' \neq b$. On an edge of F_μ, $h = 0$ and $s = f' \neq 0$. Hence s is a nowhere-zero $K \cong \mathbb{F}_2^2$-flow on all of G, which is a Tait colouring. This contradicts (6.1). □

7 The Petersen fixed-line obstruction

We now give all data needed for a finite example. The graph has vertices $0, \ldots, 9$; its edges, edge identifiers, and flow values are:

id	edge	$f(e)$	id	edge	$f(e)$
0	$0-2$	6	8	$3-5$	5
1	$0-4$	3	9	$3-8$	7
2	$0-6$	5	10	$4-5$	6
3	$1-4$	5	11	$5-9$	3
4	$1-7$	1	12	$6-8$	3
5	$1-8$	4	13	$6-9$	6
6	$2-3$	2	14	$7-9$	5
7	$2-7$	4			

(7.1)

The three incident values at vertices $0, \ldots, 9$ are, respectively,

$$\begin{gathered}(6,3,5),(5,1,4),(6,2,4),(2,5,7),(3,5,6),\\(5,6,3),(5,3,6),(1,4,5),(4,7,3),(3,6,5).\end{gathered} \tag{7.2}$$

Each triple has XOR zero, so (7.1) is a nowhere-zero $\mathbb{F}_2^3$-flow. The edge list is a labelling of the Petersen graph; its nauty-canonical graph6 encoding is

`IsP@OkWHG`.

Theorem 7.1 (two failed prescribed lines)**.** *For the flow (7.1), neither $L = \{1,2,3\}$ nor $L = \{1,6,7\}$ can be made clean by a nowhere-zero $\mathbb{F}_2^3$-flow having the same corresponding functional projection.*

Proof. For $L = \{1,2,3\}$, the kernel factor consists of edge identifiers

$$\{1,4,6,11,12\}; \tag{7.3}$$

for $L = \{1,6,7\}$, it consists of

$$\{0,4,9,10,13\}. \tag{7.4}$$

In each case the five edges are pairwise disjoint and cover all ten vertices. Contracting them in the graph (7.1), the remaining ten edges join every pair of contracted vertices exactly once. Thus $Q = K_5$.

The Petersen graph is not Tait-colourable. For a self-contained finite check, its perfect matchings in the edge indexing (7.1) are exactly

$$\begin{gathered}(0,3,8,12,14), \quad (0,4,9,10,13),\\(1,4,6,11,12), \quad (1,5,7,8,13),\\(2,3,7,9,11), \quad (2,5,6,10,14).\end{gathered} \tag{7.5}$$

Exhaustiveness follows by branching on the three edges incident with vertex 0; each initial choice has the two completions displayed in (7.5). Direct deletion shows that the complement of every matching in (7.5) is two 5-cycles. In a Tait colouring, one colour class is a perfect matching and its complement alternates the other two colours on even cycles, a contradiction.

Now K_5 is simple, has ten edges, and has girth three, so

$$10 < 4 \cdot 3.$$

Theorem 6.1 applied to (7.3) and (7.4) proves both claims. $\square$

Remark 7.2 (not a 5-CDC counterexample). The following five Eulerian edge-id sets cover every edge of (7.1) exactly twice:

$$\begin{aligned}&\{0,1,3,4,7\},\\&\{3,5,8,9,10\},\\&\{0,2,6,8,11,13\},\\&\{1,2,4,5,10,11,12,14\},\\&\{6,7,9,12,13,14\}.\end{aligned} \tag{7.6}$$

Even degree at every vertex and double coverage can both be checked directly from (7.1). Thus (7.6) is a positive 5-CDC of the example.

Remark 7.3 (the surviving existential branch). The other five lines of the displayed flow have clean lifts with fixed functional projection. A complete classification of the 64 possible binary projections on the Petersen graph finds 57 feasible projections; the failures are the zero cycle and the six ten-edge complements of the perfect matchings in (7.5). The six nonzero failures have rank five and their only dependence is their total XOR. Consequently, the three-dimensional projection space of any nowhere-zero $\mathbb{F}_2^3$-flow on the Petersen graph contains at most three of them, so at least four of its seven lines are cleanable. This finite classification is supplied as a reproducible computational audit, not needed for Theorem 7.1.

8 A cyclically four-edge-connected fixed-flow obstruction

We now show that the Petersen phenomenon cannot be removed merely by passing to the usual cyclically four-edge-connected domain. The result is stronger in the number of failed projections but remains deliberately separate from FiveCDC.

Theorem 8.1. *There is a simple nonplanar cubic graph G of order 60, with cyclic edge-connectivity at least four, and a nowhere-zero $\mathbb{F}_2^3$-flow f such that no nonzero functional projection $\mu \circ f$ has a clean fixed-projection lift. Nevertheless G has a five-cycle double cover.*

We give a deterministic construction before proving the theorem. Identify the elements of $\mathbb{F}_2^3$ with the three-bit integers $0, \ldots, 7$, using bitwise xor for addition. Begin with the order-32 graph whose graph6 record is

```
_??G@EOGG?GB_AO_g_?CP_??C??O????[O?CA?AG??CA??@???A?O??C???????G????g???@W???E???
?@c.
```

In graph6 edge order, assign the 48 values

```
3 6 2 7 5 7 2 7 5 2 1 7 2 4 6 5
6 6 5 3 6 5 5 5 6 3 6 3 6 3 5 6
3 2 4 4 1 1 6 7 2 3 1 2 6 6 2 4.
```

Their xor is zero at every vertex.

For the first block A, delete the endpoints $6, 7$ of edge 2 and apply the linear value permutation

$$T_A = (0, 4, 3, 7, 1, 5, 2, 6).$$

For the second block B, delete the endpoints $8, 9$ of edge 6 and apply

$$T_B = (0, 7, 3, 4, 5, 2, 6, 1).$$

Each tuple lists the images of $0, \ldots, 7$. The sorted ports, displayed as old vertex/value pairs before and after the maps, are

	before	after
A	$(2,6),(13,4),(15,6),(25,4)$	$(2,2),(13,1),(15,2),(25,1)$
B	$(0,7),(2,7),(3,5),(11,5)$	$(0,1),(2,1),(3,2),(11,2)$.

Relabel the retained vertices of A increasingly by $0, \ldots, 29$ and those of B by $30, \ldots, 59$. Match port indices by

$$(A_0, B_2), (A_1, B_0), (A_2, B_3), (A_3, B_1). \tag{8.1}$$

The four resulting edges and values are

$$(2,33) : 2, \quad (11,30) : 1, \quad (13,39) : 2, \quad (23,32) : 1. \tag{8.2}$$

Thus the construction is completely specified by the displayed data.

Lemma 8.2 (local obstruction)**.** *Cut the four joining edges in (8.1) into semiedges. If the local two-cycle formula is infeasible on either pole for a functional μ, after imposing the component equations only on factor components which touch no factor-valued semiedge, then no completion has a global clean fixed-projection lift for μ.*

Proof. The restriction of a global binary cycle to a pole has even incidence at every retained vertex, and hence is a binary pole cycle. Global coverage of the kernel factor restricts to coverage of every proper edge and semiedge occurrence. A local factor component touching no factor-valued semiedge is also a complete global factor component contained in the pole. Its product-parity equation (4.3) is therefore among the global cleanliness equations. A global solution would consequently restrict to a local one. □

Proof of Theorem 8.1. The two exact local bad-functional profiles, after the displayed linear maps, are

$$\mathrm{bad}(A) = \{5, 6, 7\}, \qquad \mathrm{bad}(B) = \{1, 2, 3, 4, 5\}. \tag{8.3}$$

Their union contains all seven nonzero functionals. Lemma 8.2 therefore proves the fixed-flow assertion.

For completeness, the finite proof of (8.3) is as follows. Each pole has 43 proper edges, four semiedges, and binary pole-cycle dimension 17. For each functional, enumerate all 2^{17} first cycles p. With p fixed, write the second cycle q in a computed cycle-space basis. Coverage forces $q(e) = 1$ whenever a kernel-factor edge is absent from p, while every closed-component equation becomes

$$\sum_{e \in \delta(X) \colon p(e)=1} q(e) = 0. \tag{8.4}$$

These are linear equations over $\mathbb{F}_2$, decided by ordinary Gaussian elimination. The companion checker performs this exhaustive calculation and obtains exactly (8.3).

The same program reconstructs G, checks 90 distinct edges and degree three, nonzero flow values and conservation, and exhausts all

$$\binom{90}{1} + \binom{90}{2} + \binom{90}{3} = 121575 \tag{8.5}$$

edge subsets. None separates two cyclic components. Nauty `planarg` independently classifies the displayed graph as nonplanar.

Finally, in graph6 edge order, the following five-bit weight-two labels give a five-cycle double cover:

```
3 20 5 20 17 9 5 12 12 9 12 5 5 6 5
3 24 5 10 9 17 24 12 20 9 17 24 18 9 17
3 18 24 10 24 24 10 18 9 17 10 18 24 9 9
6 6 10 9 6 3 5 12 20 18 10 24 17 3 3
17 18 5 5 10 17 9 24 20 12 12 20 24 18 18
10 24 3 3 24 17 9 18 10 24 9 17 10 18 24.
```

Weight two gives exact double coverage. Xor zero at each vertex gives even degree in every coordinate. The checker verifies both conditions directly, completing the proof. □

Remark 8.3. The proof of (8.3) is computer-assisted but solver-independent. It does not claim that a human manually inspected roughly two million cycle choices. Rather, the short exhaustive program and the reduction to linear systems constitute the finite certificate. The exact graph6 records, canonicalization command, construction tables, and checker output are preserved in the companion note.

9 Reproducibility and logical status

The companion directory

`search/fano-two-cycle-petersen-countermodel-20260726/`

contains the canonical graph, the construction table, a semantic checker, two ordinary CNF instances, and DRAT proofs. From the repository root, run

```
python3 search/fano-two-cycle-petersen-countermodel-20260726/\
  independent_checker.py
python3 search/fano-two-cycle-petersen-countermodel-20260726/\
  verify_package.py
```

The first program independently enumerates the 64 binary cycles and all 4096 ordered pairs (p, q) for each line. For each failed line, 960 pairs cover the five matching edges and none satisfies all five product-parity equations. The second program reconstructs two 65-variable, 210-clause CNFs and checks their DRAT certificates when `drat-trim` is available.

These computations are redundant for the mathematical nonexistence proof: Theorem 7.1 follows from the displayed flow, the two matching contractions, the elementary edge-count in K_5, and the six-matchings proof that Petersen is non-Tait. The computational package serves to catch transcription errors and to support the broader classification in the last remark.

The unbounded construction is replayed separately by

```
python3 scratch/verify_unbounded_fano_line_components.py
```

using only the Python standard library. It constructs depths zero through three, checks the graph and spanning-tree data, every flow and five-cover equation, exact line-component counts $1, 2, 3, 4$, and zero defect in every displayed component. The universal assertion in Theorem 5.2 rests on the displayed induction, not on exhaustive flow enumeration.

The order-60 theorem is replayed without a SAT solver by

```
c++ -O3 -std=c++20 \
  scratch/verify_fano_cyclic4_allseven_order60.cpp \
  -o /tmp/verify-fano-order60
/tmp/verify-fano-order60
```

The checker reconstructs both poles and the glued graph, derives the two profiles in (8.3), exhausts (8.5), and checks the displayed five-cover. The companion human note is `scratch/fano-cyclic4-allseven-order60-20260728.md`.

10 What is and is not new

The literature search for this draft covered the primary sources cited above and keyword/full-text searches through 28 July 2026. We did not find the combined-image theorem, the potential (3.8), the exact fixed-projection normal form (4.1)–(4.3), or the simple-contraction obstruction in those sources. Targeted searches through 28 July 2026 also did not locate the unbounded kernel-line component theorem or the cyclically four-edge-connected fixed-flow obstruction of Theorem 8.1. We therefore regard these as *plausibly new*, not as established priority claims.

The broader setting is not new: Fano colourings, line counts, their matching interpretations, and constrained $\mathbb{F}_2^2$-flows on contractions all have substantial prior literature. Most importantly, Mkrtchyan's distinct Petersen obstruction means that the appearance of the Petersen graph itself carries little novelty. The potentially publishable unit is instead the following coherent package:

1. a universal first-order span identity with an explicit telescoping certificate;
2. an exact quadratic formulation explaining why that identity does not automatically integrate to a valid switch sequence; and
3. three structural limitations: a matching-contraction obstruction instantiated by a small fixed-projection counterexample, an infinite family excluding every bounded-component normal form, and a cyclically four-edge-connected flow on which all seven fixed projections fail.

This is suitable at most as a short technical note until expert review confirms the literature boundary and a human author independently verifies the proofs. It is not a resolution, a claimed breakthrough on 5-CDC, or evidence against 5-CDC: the example positively satisfies the conjectured conclusion.

Acknowledgement and responsible authorship note

This draft was produced in an autonomous-conjecture-resolution project directed by the user. AI assistance was substantive at every stage, including conjecture formation, proof discovery, computation, literature triage, and writing. Before any public submission, a human author should rederive the arguments, check the primary literature directly, rerun the artifacts in a clean environment, decide whether the contribution merits dissemination, and take responsibility for the final text. The model provider and model are tools, not authors.

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